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Multicorrelator Signal Tracking and Signal Quality Monitoring for GNSS with Extended Kalman Filter

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Abstract—GNSS satellite signals may present anomalies that degrade the positioning performance of GNSS receivers. Signal Quality Monitoring (SQM) is normally used to detect and to characterize these anomalies. This is required for the GNSS operators and integrity services to determine when a satellite should be considered as faulty and draw conclusions about the type of the fault. In this paper, we present a new SQM algorithm that tracks the GNSS signal and possible channel deformations by using a novel Extended Kalman Filter (EKF). The EKF is designed so that the measurement update is performed in post-correlation and with multiple correlators. After the estimation of the channel response we add a detection step to determine if the channel deviates from the nominal signal transmission scenario. Results suggest that the performance of the delay estimation with the proposed EKF structure outperforms the classical DLL estimation, especially in the presence of distortions. Furthermore, it can reliably detect anomalous signal deformations as specified by ICAO threat model.

TABLE OF CONTENTS

1	INTRODUCTION	1
2	SIGNAL MODEL	2
3	PROPOSED ALGORITHM	3
4	SIMULATION RESULTS	5
5	SUMMARY	8
	REFERENCES	9
	BIOGRAPHY	10

1. INTRODUCTION

Global Navigation Satellite Systems (GNSS) signal anomalies at the satellite transmitter level, such as analog distortions, may arise due to multiple reasons and can introduce a non-negligible error in the line-of-sight (LOS) delay estimation [1, 2]. In the light of future aircraft precision landing requirements, signal quality needs to be constantly monitored.

As soon as a transmit signal anomaly is observed, the satellite needs to be marked as faulty (*signal anomaly detection*). Such faults have similar effects as an additional filter structure (i.e., channel) that distorts the ideal signal. The current signal deformation model (ICAO Threat Model [3]) restricts itself on the type of anomalies it can describe. Data from high-gain

antenna measurements have been used to better characterize GNSS signal deformations and to assess the signal quality with very high resolution [4, 5] (*signal anomaly characterization*). However, permanent recording of such data is not feasible in practice, which is why temporary anomalies may pass unnoticed or at least cannot be characterized. A low-cost signal quality monitor operating in real-time could be used to trigger high-gain measurements in case of a detected anomaly, but should also already provide an approximate qualitative picture of the signal deformation at the moment the anomaly occurs. A first characterization could be given, for instance, in terms of a non-ideal impulse response.

Since most GNSS employ pseudonoise (PN) sequences, multicorrelator architectures have been used successfully for the signal anomaly detection problem [6–8]. This corresponds to sampling of the cross-correlation between ideal replica and received signal at a finite number of points in delay domain around the correlation peak. In these works, sophisticated decision metrics based on all correlator outputs have been formulated to raise an alarm reliably in case of an anomaly. However, the detection metrics alone can only provide a limited characterization of the anomaly. Moreover, synchronization in the above works is achieved traditionally using one pair of correlators and ignoring the remaining “slave” correlators.

However, all correlator outputs do provide valuable information for synchronization and signal anomaly characterization, since they can be used for accurate and robust channel parameter tracking. Well-known examples for using multiple correlators are the “very early/very late” slave correlators [9] for robust tracking of offset carrier signals or the multipath estimating delay-locked loop (MEDLL) [10, 11] that uses one tracking loop per path to track and subtract multipath signals. The first systematic approach to exploit multiple correlators for parameter estimation without redundancy can be found in [12], which has inspired many works on GNSS parameter estimation/tracking in multipath channels [13–16]. In those works, a (quasi-)static correlator bank is used, whose output can serve to estimate or track a large number of channel parameters such as path amplitudes and delays optimally in a maximum likelihood (ML) [13] or Bayesian sense [16], or using hybrid optimality criteria [14, 15]. Since correlator banks can only cover a limited range of possible delays (for instance, an interval of two chip durations), it is important to adjust the correlator bank according to the most recent LOS delay estimate, or otherwise the measurement will suffer from correlation loss as soon as the true LOS delay leaves the

delay range covered by the correlator bank. This feedback effect is not considered in the available literature on optimal multicorrelator LOS tracking to our knowledge.

Another issue that has to be considered when aiming at permanent real-time signal characterization is model complexity. Especially, nonlinear parameter estimation in many dimensions should be avoided when possible, since this requires high computational effort. Therefore, a system model should be pursued which has very few nonlinear parameter dependencies but still represents the channel adequately. A promising framework was proposed in [17], where an Extended Kalman Filter (EKF) operating at the sampling rate in pre-correlation domain was used to track the parameters of a tapped delay-line channel impulse response, the shortest delay being the only parameter which introduces nonlinearity.

In this work, we propose a multicorrelator EKF framework in post-correlation domain for joint tracking of LOS delay and the coefficients of a tapped delay-line channel, taking into account arbitrary bandwidth of the nominal signal. The EKF provides a solid and extensively investigated framework with which the correlator bank outputs can be processed to provide quasi-optimal estimates for the modeled time-variant states. In the anomaly-free (nominal) case, all estimated coefficients except the one corresponding to the LOS have a near-zero magnitude. In case of a satellite signal anomaly, the deviation of the channel coefficient estimates from their nominal values can be used to raise a detection alarm. The EKF filter structure estimates the sudden changes in the received signal, matches the deformation by estimating the non-zero channel coefficients and thus negates the effect of the fault. Thus, the algorithm can not only be used for signal anomaly detection but also to characterize the satellite fault quantitatively while keeping unbiased track of the signal, contrary to state-of-the-art detectors. Additionally, the proposed algorithm is lightweight enough to be implemented for true real-time signal quality monitoring. We believe that this is a significant contribution, because literature does not provide a multicorrelator-based methodology which allows combined tracking, characterization and monitoring of GNSS signals and signal deformations suited for operation in real-time. A second element of novelty is that we consider the feedback effect which the time-variant LOS estimate has on the adjustment of the correlator bank.

2. SIGNAL MODEL

A GNSS satellite generates a periodic PN signal

$$s(t) \equiv s(t - mT) \quad \forall m \in \mathbb{Z} \quad (1)$$

with autocorrelation function (ACF)

$$\Phi_{ss}(\tau) = \int_T s(t)s(t + \tau)dt, \quad (2)$$

where we denote the signal period by T and let $\Phi_{ss}(0) = 1$. The signal has a chipping rate of $1/T_c$ and is bandlimited to one-sided bandwidth $B = \lambda/T_c$, where $\lambda \in \mathbb{N}$. If transmission and channel were distortionless, the signal observed by a receiver with radial distance $r(t)$ with respect to the satellite would only experience a range-dependent propagation delay and additive white Gaussian noise (AWGN), making the received signal a scaled and noisy version of $s(t - \tau^{(0)}(t))$ with $\tau^{(0)}(t) \propto r(t) > 0$ and $\eta(t)$ having zero mean and unit variance. However, due to manufacturing imperfections or

aging effects in the satellite payload components, the purity of the signal may be affected. Analog distortion effects can be modelled as a filtering operation on the ideal signal. Therefore, the received signal can be expressed as

$$y(t) = \int_{-\infty}^{\infty} s(t - \tau - \tau^{(0)}(t))h(t, \tau) d\tau + \eta(t). \quad (3)$$

At this point, it is important to remember that the satellite signal transmission is a causal system, i.e., a system where the current signal $y(t)$ only depends on the past and current inputs but not on the future ones. Thus, the integral in (3) may be taken over all nonnegative delays $\tau \geq 0$. Additionally, due to the nature of the pseudorandom signals, signal contributions with a delay $\tau > T_c$, (i.e., larger than one chip period) are decorrelated and treated as part of the noise term $\eta(t)$. Therefore, we can further simplify (3) to

$$y(t) = \int_0^{T_c} s(t - \tau - \tau^{(0)}(t))h(t, \tau) d\tau + \eta(t) \quad (4)$$

We approximate the channel by a sequence of Dirac functions arranged on a tapped delay-line

$$h(t, \tau) \approx \sum_{\ell=0}^L h^{(\ell)}(t)\delta(\tau - \ell T_s) \quad (5)$$

with time-variant weights $h^{(\ell)}(t)$, which is a valid assumption for small spacing $T_s \ll T_c$ and L satisfying $LT_s = T_c$. Inserting (5) into (4), we get

$$\begin{aligned} y(t) &\approx \int_0^{T_c} \sum_{\ell=0}^L h^{(\ell)}(t)\delta(\tau - \ell T_s)s(t - \tau - \tau^{(0)}(t)) d\tau \\ &\quad + \eta(t) \\ &\approx \sum_{\ell=0}^L h_k^{(\ell)} \int_0^{T_c} \delta(\tau - \ell T_s)s(t - \tau - \tau_k^{(0)}) d\tau + \eta(t) \\ &= \sum_{\ell=0}^L h_k^{(\ell)} s(t - \tau_k^{(0)} - \ell T_s) + \eta(t). \end{aligned} \quad (6)$$

Since the channel coherence time is typically larger than T , the last approximation in the above equation is that the LOS delay and the channel weights are blockwise constant, $\tau_k^{(0)} \triangleq \tau^{(0)}(kT)$ and $h_k^{(\ell)} \triangleq h^{(\ell)}(kT)$ for $k = \lfloor t/T \rfloor$ and $\ell = 0, \dots, L$.

As a next processing step, the received signal is filtered with one-sided bandwidth B and sampled at the rate $\frac{1}{T_s} = 2B$, which is the Nyquist rate. The design parameter λ can be chosen sufficiently high such as to satisfy $T_s \ll T_c$ satisfied. The number of observable channel coefficients is then $L = \frac{T_c}{T_s} = 2\lambda$. Sampling the signal in (6) beginning at time $t = 0$ leads to

$$\begin{aligned} y_k[n] &\triangleq y((kN + n)T_s) \\ &= \sum_{\ell=0}^L h_k^{(\ell)} s((kN + n)T_s - \tau_k^{(0)} - \ell T_s) \\ &\quad + \eta((kN + n)T_s) \\ &= \sum_{\ell=0}^L h_k^{(\ell)} s((n - \ell)T_s - \tau_k^{(0)}) + \eta((kN + n)T_s), \end{aligned} \quad (7)$$

where we used the periodicity of the PN sequence $s(t)$. Collecting periods $k = 0, 1, \dots$ and stacking N samples per period in one column vector leads to

$$\mathbf{y}_k = [y_k[1], \dots, y_k[N]]^T \in \mathbb{R}^{N \times 1}, \quad k = 0, 1, \dots \quad (8)$$

Thus, our discrete signal can be written in matrix-vector notation as

$$\mathbf{y}_k = \mathbf{H}_k \mathbf{s}(\tau_k^{(0)}) + \boldsymbol{\eta}_k, \quad (9)$$

with the ideal signal

$$\mathbf{s}(\tau_k^{(0)}) = \begin{pmatrix} s(T_s - \tau_k^{(0)}) \\ s(2T_s - \tau_k^{(0)}) \\ \vdots \\ s(NT_s - \tau_k^{(0)}) \end{pmatrix} \in \mathbb{R}^{N \times 1}$$

and white noise with i.i.d. standard normally distributed entries

$$\boldsymbol{\eta}_k = [\eta(T_s + kT), \dots, \eta(NT_s + kT)]^T. \quad (10)$$

The channel is expressed by a convolution matrix with circulant Toeplitz structure

$$\mathbf{H}_k = \sum_{\ell=0}^L h_k^{(\ell)} \mathbf{Z}^\ell \in \mathbb{R}^{N \times N}, \quad (11)$$

$\mathbf{Z}$ being the polynomial basis of $N \times N$ circulant matrices

$$\mathbf{Z} = \begin{pmatrix} \mathbf{0}^T & \mathbf{1} \\ \mathbf{I} & \mathbf{0} \end{pmatrix} \in \mathbb{R}^{N \times N}. \quad (12)$$

The unknown time-varying signal and channel parameters are summarized by the state vector

$$\mathbf{x}_k = [\tau_k^{(0)}, h_k^{(0)}, h_k^{(1)}, \dots, h_k^{(L)}]^T \in \mathbb{R}^{(L+2) \times 1}. \quad (13)$$

3. PROPOSED ALGORITHM

Measurement Model - Multicorrelator Bank

The number of data collected per period N proves to be computationally challenging. Thus, we need an efficient representation of the incoming signal without loss of information. In order to do so we employ a projection of the observed signal to a much smaller space, thus reducing the subsequent processing requirements. A variant of these projections is employing a bank of signal-matched correlators and building the temporal cross-correlation function of the incoming signal with a local set of time shifted replicas [18]

$$\begin{pmatrix} \mathbf{s}^T(\hat{\tau}_k^{(-q)}) \\ \mathbf{s}^T(\hat{\tau}_k^{(q-1)}) \\ \vdots \\ \mathbf{s}^T(\hat{\tau}_k^{(0)}) \\ \vdots \\ \mathbf{s}^T(\hat{\tau}_k^{(q-1)}) \\ \mathbf{s}^T(\hat{\tau}_k^{(q)}) \end{pmatrix} \mathbf{y}_k \in \mathbb{R}^{P \times 1}, \quad (14)$$

where $\mathbf{s}(\cdot) \in \mathbb{R}^{N \times 1}$ is the ideal bandlimited signal, $\hat{\tau}_k^{(q)}$ represents the temporal delay of the q -th local PN replica and $P = 2q + 1$ is the total number of local replicas. Through this operation, we project the signal from a $\mathbb{R}^{N \times 1}$ space to a much smaller $\mathbb{R}^{P \times 1}$ space. In order to minimize the loss of information due to the dimensional collapse, we assume that the correlator bank's middle/prompt replica is roughly synchronized with the LOS signal, i.e., $|\hat{\tau}_k^{(0)} - \tau_k^{(0)}| \leq T_c/2$. Additionally, we place the remaining $2q$ correlators symmetrically and equidistantly around the prompt correlator within $\pm T_c$ using a spacing of $\Delta = \frac{T_c}{q}$. Thus we have

$$\hat{\tau}_k^{(\pm i)} - \hat{\tau}_k^{(0)} = \pm i\Delta, \quad i = 1, \dots, q. \quad (15)$$

This simplifies the parameters which describe the correlator bank

$$\mathbf{Q}^T(\hat{\tau}_k^{(0)}) \triangleq \begin{pmatrix} \mathbf{s}^T(\hat{\tau}_k^{(0)} - q\Delta) \\ \mathbf{s}^T(\hat{\tau}_k^{(0)} - (q-1)\Delta) \\ \vdots \\ \mathbf{s}^T(\hat{\tau}_k^{(0)}) \\ \vdots \\ \mathbf{s}^T(\hat{\tau}_k^{(0)} + (q-1)\Delta) \\ \mathbf{s}^T(\hat{\tau}_k^{(0)} + q\Delta) \end{pmatrix} \in \mathbb{R}^{P \times N}, \quad (16)$$

which is now determined only by Δ , which is the temporal spacing of adjacent correlators, and $\hat{\tau}_k^{(0)}$, which is the estimated delay of the LOS signal. The result of the cross-correlation yields the post-correlation signal

$$\begin{aligned} \mathbf{z}_k &= \mathbf{Q}^T(\hat{\tau}_k^{(0)}) \mathbf{y}_k \\ &= \mathbf{Q}^T(\hat{\tau}_k^{(0)}) \mathbf{H}_k \mathbf{s}(\tau_k^{(0)}) + \mathbf{Q}^T(\hat{\tau}_k^{(0)}) \boldsymbol{\eta}_k \end{aligned} \quad (17)$$

for $k = 0, 1, \dots$

Extended Kalman Filter Description

The received signal $\mathbf{y}_k$ in the absence of noise can be fully described given the state $\mathbf{x}_k \in \mathbb{R}^{L+2}$ from (13). Thus, given a multicorrelator bank, we set the correlator spacing $\Delta = T_s$, which results in a number of correlators $P = 2L + 1$. This way, we are able to align the $L + 1$ later correlators of the bank with the channel $\mathbf{H}_k$ experienced by the satellite signal $\mathbf{s}(\tau_k^{(0)})$. In the sequel, we propose a post-correlation EKF which for each $k = 0, 1, \dots$ estimates the state vector $\mathbf{x}_k$ with quasi minimum mean square error based on the measurements $\mathbf{z}_0, \dots, \mathbf{z}_k$.

The channel coefficients $h_k^{(\ell)}$ and LOS delay $\tau_k^{(0)}$ are assumed to be independent and follow a random walk dynamic model

$$\tau_k^{(0)} = \alpha_\tau \tau_{k-1}^{(0)} + u_{\tau,k} \quad (18)$$

$$h_k^{(\ell)} = \alpha_h^{(\ell)} h_{k-1}^{(\ell)} + u_{h,k}^{(\ell)} \quad \forall \ell = 0, 1, \dots, L, \quad (19)$$

where $u_{\tau,k}$ and $u_{h,k}^{(\ell)}$ are mutually independent additive white Gaussian noise processes with variances β_τ and $\beta_h^{(\ell)}$. State transition model and measurement model can be written more

compactly as

$$\mathbf{x}_k = \mathbf{A}\mathbf{x}_{k-1} + \mathbf{u}_k \quad (20)$$

$$\mathbf{z}_k = \mathbf{f}(\mathbf{x}_k, \mathbf{x}_k^-, \boldsymbol{\eta}_k) \quad (21)$$

with the state transition matrix

$$\mathbf{A} = \begin{pmatrix} \alpha_\tau & 0 & 0 & \dots & 0 \\ 0 & \alpha_{h^{(0)}} & 0 & \dots & 0 \\ 0 & 0 & \alpha_{h^{(1)}} & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 0 & \alpha_{h^{(L)}} \end{pmatrix} \quad (22)$$

and the process noise

$$\mathbf{u}_k = [u_{\tau,k}, u_{h,k}^{(0)}, u_{h,k}^{(1)}, \dots, u_{h,k}^{(L)}]^T \quad (23)$$

with process noise covariance matrix

$$\mathbf{B} \triangleq \mathbb{E}[\mathbf{u}_k \mathbf{u}_k^T] = \begin{pmatrix} \beta_\tau & 0 & 0 & \dots & 0 \\ 0 & \beta_{h^{(0)}} & 0 & \dots & 0 \\ 0 & 0 & \beta_{h^{(1)}} & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 0 & \beta_{h^{(L)}} \end{pmatrix}. \quad (24)$$

The nonlinear measurement function is given by

$$\mathbf{z}_k = \mathbf{f}(\mathbf{x}_k, \mathbf{x}_k^-, \boldsymbol{\eta}_k) = \mathbf{Q}^T(\tau_k^{(0)-}) [\mathbf{H}_k \mathbf{s}_k(\tau_k^{(0)}) + \boldsymbol{\eta}_k], \quad (25)$$

where $\tau_k^{(0)-}$ denotes the prediction of the LOS delay $\tau_k^{(0)}$. It is important to notice that at time k , the measurement $\mathbf{z}_k$ depends (nonlinearly) not only on noise and the true state $\mathbf{x}_k$, but also on the current prompt delay of the correlator bank. For instance, if the prompt delay deviates from the true LOS delay by more than $2T_c$, the measurement will contain no useful signal but colored noise only. We assume that the correlator bank's prompt delay for generation of $\mathbf{z}_k$ is driven by the best current estimate, which is the prediction $\tau_k^{(0)-}$.

Prediction Stage

In the joint delay and channel estimation and tracking problem, our state prediction model is linear as shown above. Thus, the EKF equations for the state prediction and associated covariance are

$$\mathbf{x}_k^- = \mathbf{A}\mathbf{x}_{k-1}^+ \quad (26)$$

$$\mathbf{P}_k^- = \mathbf{A}\mathbf{P}_{k-1}^+ \mathbf{A}^T + \mathbf{B}, \quad (27)$$

where $\mathbf{P}_{k-1}^+$ is the covariance associated with the EKF's state estimate at the end of the previous iteration (update). Further, the predicted measurement is defined as

$$\mathbf{z}_k^- = \mathbb{E}\{\mathbf{z}_k | \mathbf{z}_{k-1}, \dots, \mathbf{z}_0\} = \mathbb{E}\{\mathbf{z}_k | \mathbf{x}_k^-\} = \mathbf{f}(\mathbf{x}_k^-, \mathbf{x}_k^-, \mathbf{0}) \quad (28)$$

which is equivalent to the measurement function evaluated at the predicted state vector $\mathbf{x}_k^-$.

Update Stage

The residual between measurement and predicted measurement is defined as

$$\boldsymbol{\epsilon}_k = \mathbf{z}_k - \mathbf{z}_k^-. \quad (29)$$

From the residual, the updated state estimate and associated covariance are obtained as

$$\mathbf{x}_k^+ = \mathbf{x}_k^- + \mathbf{K}_k \boldsymbol{\epsilon}_k \quad (30)$$

$$\mathbf{P}_k^+ = (\mathbf{I} - \mathbf{K}_k \mathbf{F}_k) \mathbf{P}_k^- \quad (31)$$

with the Kalman gain

$$\mathbf{K}_k = \mathbf{P}_k^- \mathbf{F}_k^T \mathbf{S}_k^{-1} \in \mathbb{R}^{(L+2) \times P}, \quad (32)$$

the residual covariance matrix

$$\mathbf{S}_k = \mathbf{F}_k \mathbf{P}_k^- \mathbf{F}_k^T + \mathbf{R}_k, \quad (33)$$

and a linearization of (25) with respect to the state, evaluated at the predicted state vector, i.e.,

$$\begin{aligned} \mathbf{F}_k &\triangleq \left. \frac{\partial \mathbf{f}(\mathbf{x}_k, \mathbf{x}_k^-, \mathbf{0})}{\partial \mathbf{x}_k} \right|_{\mathbf{x}_k = \mathbf{x}_k^-} \\ &= \left(\begin{array}{c} \frac{\partial \mathbf{f}^T(\mathbf{x}_k, \mathbf{x}_k^-, \mathbf{0})}{\partial \tau_k^{(0)}} \\ \frac{\partial \mathbf{f}^T(\mathbf{x}_k, \mathbf{x}_k^-, \mathbf{0})}{\partial h_k^{(0)}} \\ \vdots \\ \frac{\partial \mathbf{f}^T(\mathbf{x}_k, \mathbf{x}_k^-, \mathbf{0})}{\partial h_k^{(L)}} \end{array} \right) \bigg|_{\mathbf{x}_k = \mathbf{x}_k^-} \in \mathbb{R}^{P \times (L+2)}. \end{aligned} \quad (34)$$

The measurement noise covariance matrix is given as

$$\begin{aligned} \mathbf{R}_k &= \mathbf{Q}^T(\tau_k^{(0)-}) \mathbb{E}[\boldsymbol{\eta}_k^T \boldsymbol{\eta}_k] \mathbf{Q}(\tau_k^{(0)-}) \\ &= \mathbf{Q}^T(\tau_k^{(0)-}) \mathbf{Q}(\tau_k^{(0)-}) \in \mathbb{R}^{P \times P}, \end{aligned} \quad (35)$$

since the pre-correlation noise $\boldsymbol{\eta}_k$ has i.i.d. **standard normally distributed entries**. Using (2) and (11), the partial derivative in (34) with respect to $\tau_k^{(0)}$ can be calculated as

$$\begin{aligned} \left. \frac{\partial \mathbf{f}(\mathbf{x}_k, \mathbf{x}_k^-, \mathbf{0})}{\partial \tau_k^{(0)}} \right|_{\mathbf{x}_k = \mathbf{x}_k^-} &= \mathbf{Q}^T(\tau_k^{(0)-}) \mathbf{H}_k \left. \frac{\partial \mathbf{s}(\tau_k^{(0)})}{\partial \tau_k^{(0)}} \right|_{\tau_k^{(0)} = \tau_k^{(0)-}} \\ &= \sum_{\ell=0}^L h_k^{(\ell)} \begin{pmatrix} \Phi'_{ss}((\ell+q)\Delta) \\ \vdots \\ \Phi'_{ss}((\ell-q)\Delta) \end{pmatrix}, \end{aligned} \quad (36)$$

where $\Phi'_{ss}(\tau) = \frac{d}{d\tau} \Phi_{ss}(\tau)$. Likewise, we calculate the partial derivative in (34) with respect to $h_k^{(\ell)}$ for $\ell = 0, \dots, L$

$$\begin{aligned} \left. \frac{\partial \mathbf{f}(\mathbf{x}_k, \mathbf{x}_k^-, \mathbf{0})}{\partial h_k^{(\ell)}} \right|_{\mathbf{x}_k = \mathbf{x}_k^-} &= \mathbf{Q}^T(\tau_k^{(0)-}) \mathbf{Z}^\ell \mathbf{s}(\tau_k^{(0)-}) \\ &= \mathbf{Q}^T(0) \mathbf{Z}^\ell \mathbf{s}(0) \\ &= \begin{pmatrix} \Phi_{ss}((q-\ell)\Delta) \\ \vdots \\ \Phi_{ss}((q+\ell)\Delta) \end{pmatrix}. \end{aligned} \quad (37)$$

Finally, the measurement noise covariance in (35) can be simplified as

$$\begin{aligned} \mathbf{R}_k &= \mathbf{Q}^T(\tau_k^{(0)-})\mathbf{Q}(\tau_k^{(0)-}) \\ &= \mathbf{Q}^T(0)\mathbf{Q}(0) \\ &= \begin{pmatrix} \Phi_{ss}(0) & \dots & \Phi_{ss}(2q\Delta) \\ \vdots & \ddots & \vdots \\ \Phi_{ss}(2q\Delta) & \dots & \Phi_{ss}(0) \end{pmatrix} \triangleq \mathbf{R} = \text{const.} \end{aligned} \quad (38)$$

It is worth noting that the measurement covariance does not depend on the prompt delay, hence is time-invariant. Moreover, the $L + 1$ latter columns of $\mathbf{F}_k$ are time-invariant, while its first column can be expressed as a weighted sum of vectors with fixed vectors and time-variant weights. These simplifications are essentially a consequence of the relationship $\mathbf{s}^T(\tau)\mathbf{s}(\tau + n\Delta) = \Phi_{ss}(n\Delta)$ for any $n \in \mathbb{Z}$, $\tau \in \mathbb{R}$. Moreover, note that the post-correlation measurement noise $\mathbf{Q}^T(\tau_k^{(0)-})\boldsymbol{\eta}_k$ does not have uncorrelated entries despite $\boldsymbol{\eta}_k$ having i.i.d. entries. Due to multiplication with the correlator bank, the measurement noise is colored with covariance $\mathbf{R}$.

Signal Anomaly Detectors

The Extended Kalman Filter (EKF) of this work has been designed to estimate the LOS time delay and the channel that the nominal signal experiences. Since we expect our EKF to match the channel response, a fault or signal anomalies detector cannot be based on the EKF. Instead we build an additional module that will perform a test to determine deviations from nominal signal transmission. The anomaly detector uses in each iteration the current a posteriori estimate $\mathbf{x}_k^+$ and covariance $\mathbf{P}_k^+$ provided by the EKF (Figure 1) and determines if an anomaly is present in the channel by comparing the current channel response with the nominal expected channel. The nominal channel response that constitutes our

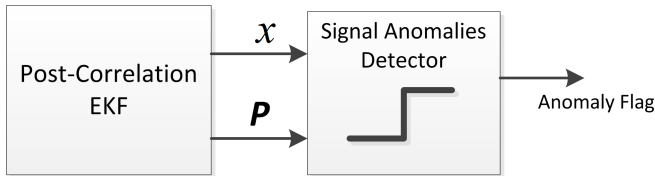


Figure 1. EKF and anomaly detector

null hypothesis can be easily defined as

$$\mathbf{x}_{k,\text{nom}} = [\tau_k^{(0)}, h_k^{(0)}, 0, \dots, 0]^T \quad (39)$$

Notice that we allow the time delay $\tau_k^{(0)}$ and LOS amplitude $h_k^{(0)}$ to have any value. Therefore, we will focus only on the coefficients from $h_k^{(1)}$ to $h_k^{(L)}$. A trimmed current and nominal state vector would be then

$$\mathbf{x}_k^* = [h_k^{(1)}, \dots, h_k^{(L)}]^T \quad (40)$$

$$\mathbf{x}_{\text{nom}}^* = [0, \dots, 0]^T \in \mathbb{R}^{L \times 1} \quad (41)$$

We will use also the current time trimmed predicted covariance matrix as $\mathbf{P}_k^{*+}$, which contains the columns and rows referred to the trimmed predicted state vector. A test

statistic can therefore be built by using the weighted residuals between the current state and nominal expected state:

$$q^2 = (\mathbf{x}_k^{*+} - \mathbf{x}_{\text{nom}}^*)^T (\mathbf{P}_k^{*+})^{-1} (\mathbf{x}_k^{*+} - \mathbf{x}_{\text{nom}}^*). \quad (42)$$

In the absence of channel anomalies, the test q^2 follows a centrally distributed chi-squared distribution with L degrees of freedom. The test statistic is compared to a threshold T to determine if the channel presents some anomaly:

$$\text{If } \begin{cases} q^2 \leq T & : \text{No anomaly} \\ q^2 > T & : \text{Anomaly Detected} \end{cases} \quad (43)$$

A suitable threshold T can be obtained from the inverse chi-squared distribution for the given degrees of freedom L and the desired probability of false alarm P_{fa} .

4. SIMULATION RESULTS

In order to test the capabilities of the algorithm, proposed in section (3), we carried out several computer simulations to demonstrate its LOS delay and channel coefficients estimation capabilities. For the following simulation scenarios investigated, we choose $\lambda = 4$. Therefore, the nominal signal has single-sided bandwidth $B = \lambda/T_c = 4.092\text{MHz}$, is sampled with the Nyquist frequency frequency $f_s = \frac{2\rho\lambda}{T_c} = 8.184\text{MHz}$ and subsequently cross-correlated with a correlator bank containing a total of $P = 2L + 1 = 17$ equispaced correlators with a spacing of $\Delta = \frac{2T_c}{L} = 0.125 T_c$. The nominal carrier-to-noise density is set to 54.7 dB-Hz. The initial delay estimation error is $\tau_0^{(0)} - \tau_0^{(0)+} = 0.0 T_c$, modeling stable tracking behavior, and the relative amplitude error is $\frac{h_0^{(0)} - h_0^{(0)+}}{h_0^{(0)}} = 0.1 = 10\%$. The associated standard deviation of the initial guess is $0.5 T_c$ for the LOS delay and 0.1 for all amplitudes. Hereafter, the ground truth delay and amplitude of the LOS signal perform a random walk as defined in Equations (18-24) with $\mathbf{A} = \mathbf{I}$, $\beta_\tau = 0.001 T_c$ and $\beta_h^{(\ell)} = 0.01$ for $\ell = 0, \dots, L$. These variances are the same as used by the EKF. For comparison, a traditional DLL with an early-late spacing of $0.25 T_c$ and a loop bandwidth of 0.5 Hz has been implemented and tested for the same scenario.

Nominal LOS Random Walk

The first simulation (Figure 2) aims at studying the nominal behavior of both algorithms. The DLL oscillates around the real delay and eventually converges to the real LOS delay value. On the other side, the proposed algorithm takes advantage of prior sample knowledge and converges faster and with a smaller variance for the real LOS delay estimation.

Figure 3 illustrates the estimation and nominal case tracking behavior of LOS channel coefficient of the proposed algorithm. We can see that the proposed algorithm follows the evolution of the LOS amplitude very well and with a very small variance.

Figure 4 shows the interpolated channel impulse response derived from the estimated state vector of the proposed algorithm. Specifically, the LOS channel coefficient and its respective delay is dominant. The remaining channel coefficients are zero mean and have Gaussian noise-like characteristics as expected, since only the LOS is present in the signal.

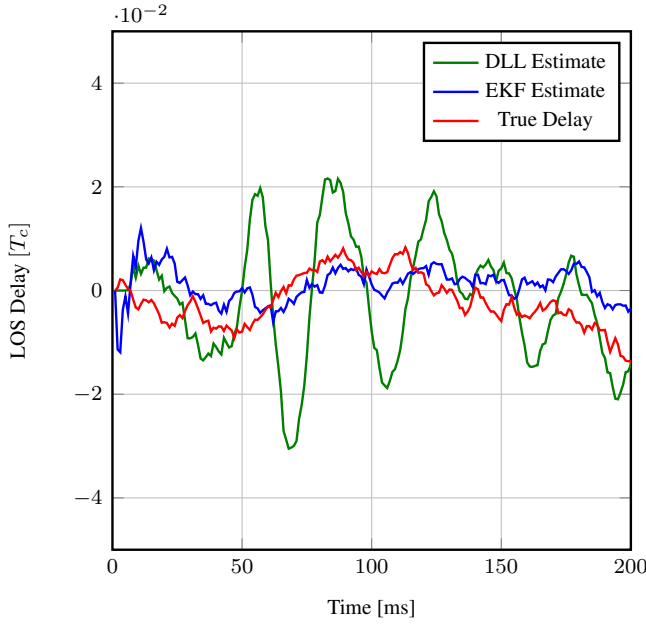


Figure 2. LOS delay estimation comparison between proposed EKF algorithm and conventional DLL for the nominal case.

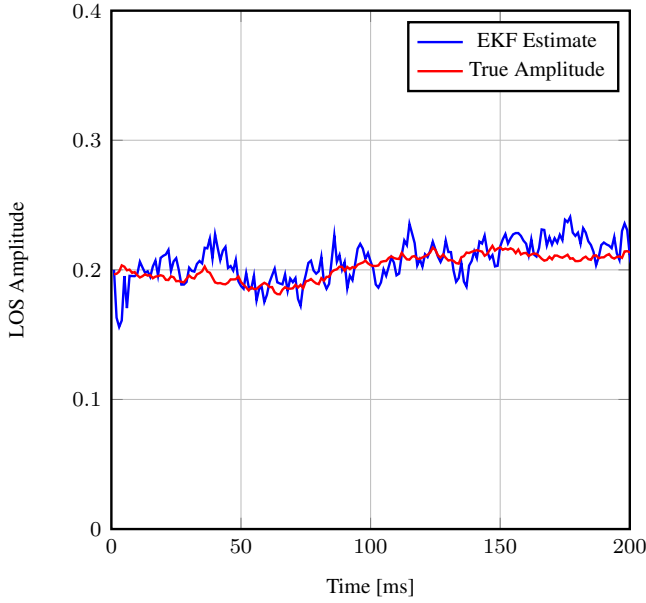


Figure 3. LOS amplitude estimation by the proposed EKF algorithm for the nominal case.

Secondary Signal Replica Delay Sweep

The second simulation involves two signal components. The LOS parameters behave as in the previous simulation. The second signal replica has a constant multipath to signal ratio (MSR) of -6dB and a time-varying delay $\tau_k^{(1)} \in [0, 1.25 T_c]$. The initial relative delay between the two signal components is $\tau_0^{(1)} - \tau_0^{(0)} = 1.25 T_c$, and subsequently decreases in a quadratic rate until $\tau_{100}^{(1)} - \tau_{100}^{(0)} = 0 T_c$ after 100 ms and finally increases again quadratically until after 200 ms $\tau_{200}^{(1)} - \tau_{200}^{(0)} = 1.25 T_c$.

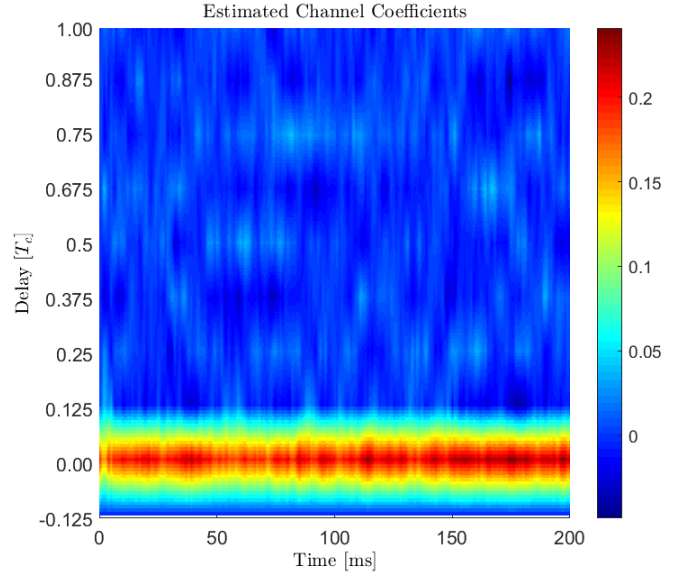


Figure 4. Interpolated channel impulse response for the nominal case.

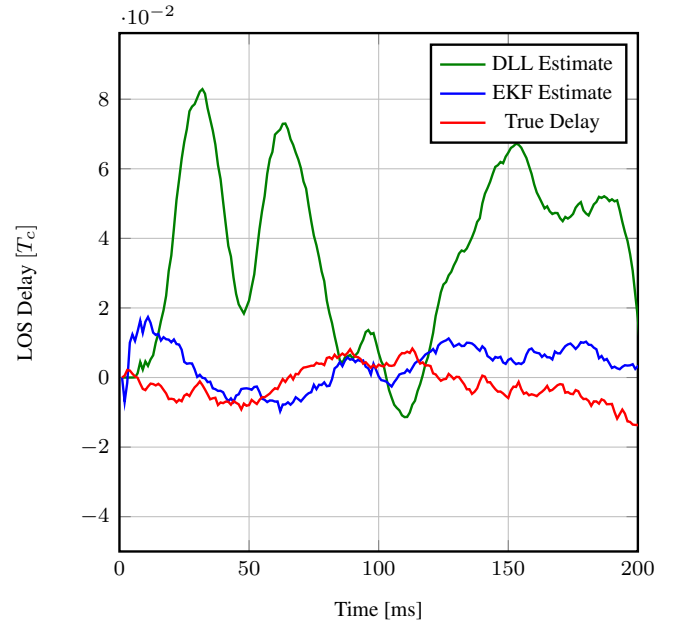


Figure 5. LOS delay estimation comparison between proposed EKF algorithm and conventional DLL for a signal replica sweep.

As seen in Figure 5, the DLL is not affected as long as the relative delay of the secondary replica is larger than one PN chip (T_c) or nearly zero and thus causes negligible deformations. On the other side, the DLL LOS delay estimation is severely affected when the second signal replica is near the late correlator tap of the DLL, i.e., the relative delay is on the order of $\frac{T_c}{2}$. The proposed EKF based algorithm remains not only unaffected by the sweeping second replica but also is able to estimate its time varying channel coefficient, as indicated in Figure 6.

As seen in the interpolated channel, the secondary replica

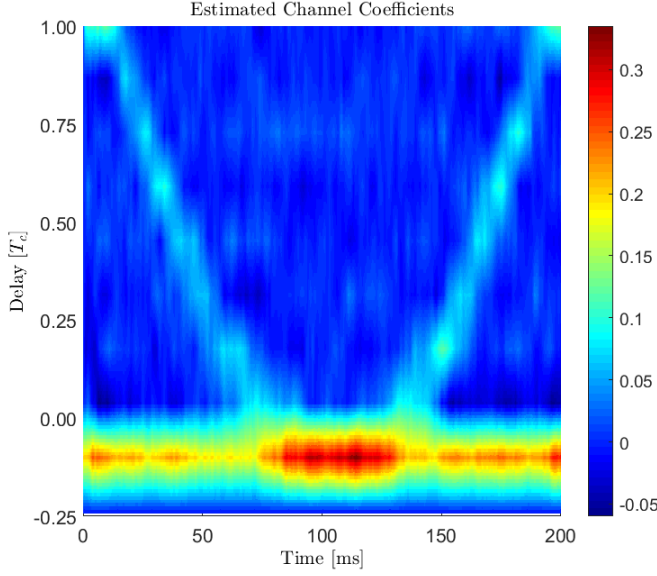


Figure 6. Interpolated channel impulse response with a second temporally sweeping signal replica.

sweeps in delay between correlators (Figure 6) and its presence it visible. As a matter of a fact, the proposed algorithm is able to correctly predict the amplitude of the second signal replica when its delay crosses that of a correlator in the bank.

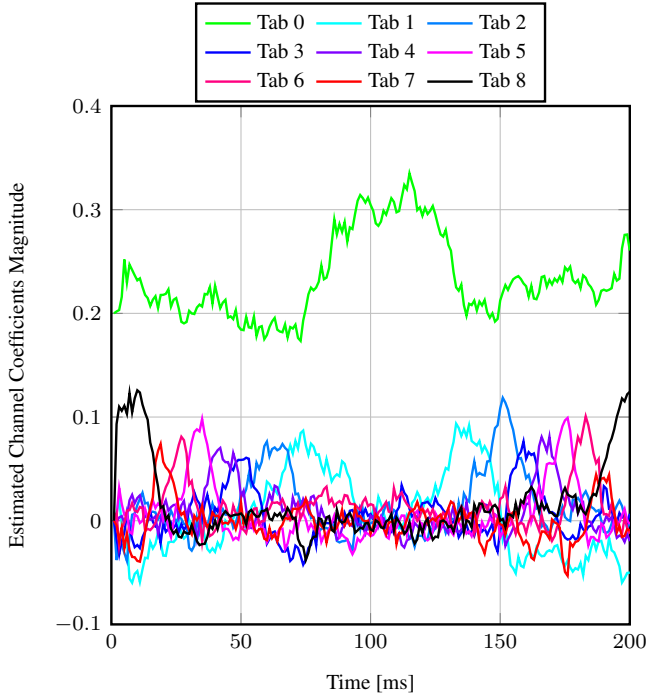


Figure 7. Channel estimation coefficients for a temporally sweeping signal replica.

Figure 7 illustrates that the maximum value of the time varying channel coefficient has an amplitude of about half of the LOS when it crosses over a correlator. The rest of the time, when the delay of the signal replica is situated

between two correlators in the temporal domain, it is being estimated as a composite signal of the correlators that are placed around it. It is interesting to note that when the relative delay becomes too small, the secondary replica is being added constructively on the LOS and increases its estimated amplitude by its magnitude.

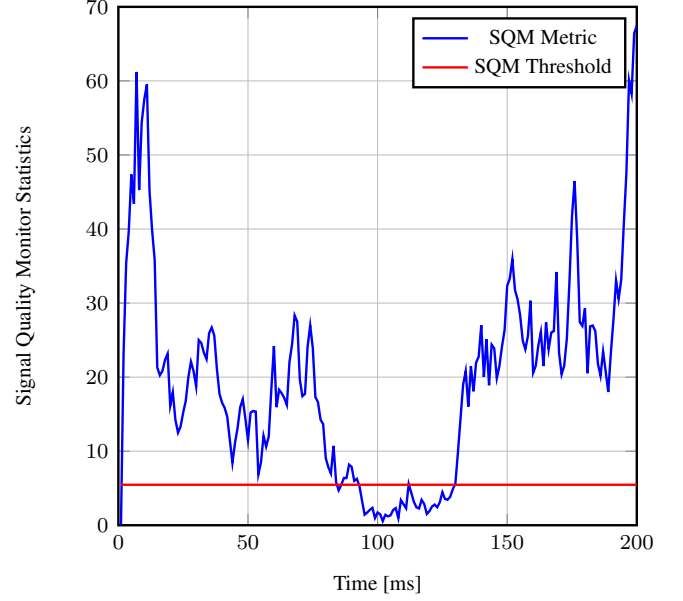


Figure 8. Signal Quality Monitoring (SQM) for a temporally sweeping signal replica.

Finally, Figure 8 shows the detection metrics for the current scenario. Since the sweeping signal replica is has a delay of over one T_c at the start of the simulation, the SQM metric does not cross the threshold bound. However, as soon as its delay gets smaller than one T_c , the SQM metric crosses the bound and an alarm is triggered. Finally, for a relative delay smaller than $0.125 T_c$, the SQM metric goes below the threshold bound since for the chosen bandwidth B the algorithm cannot distinguish between the two signals and the resolution limit of the algorithm has been reached.

ICAO Analog Distortion Model

The ICAO threat model was developed to approximate the signal deformations observed in the SV19 signal faults [19, 20]. This model describes two kinds of faults, that is digital and analog distortions. The digital distortions describe PN pulse jitter and the analog distortions model failures of the power amplifiers on the satellite. The latter fault is modeled as a filtering operation for the band limited PN sequence with the impulse response of an IIR lowpass filter. Thus, it induces a damped oscillation at every chip transition as seen in Figure 9. In the present simulation, we simulate an ideal bandlimited signal for 50ms. Thereafter it is being filtered by an IIR filter with a resonant frequency of $f_d = 3\text{ MHz}$ and a damping ratio $\sigma = 8.8\text{ Np}/\mu\text{s}$, which is equivalent to a sudden analog distortion fault at the satellite.

In Figure 10, we see that the DLL suffers from the sudden distortion occurrence and a bias in the delay estimation can be observed. The proposed algorithm also suffers a bias from the analog distortion but smaller in magnitude.

The interpolated estimated channel impulse response can be seen in Figure 11. For the first 50ms of the simulation, the

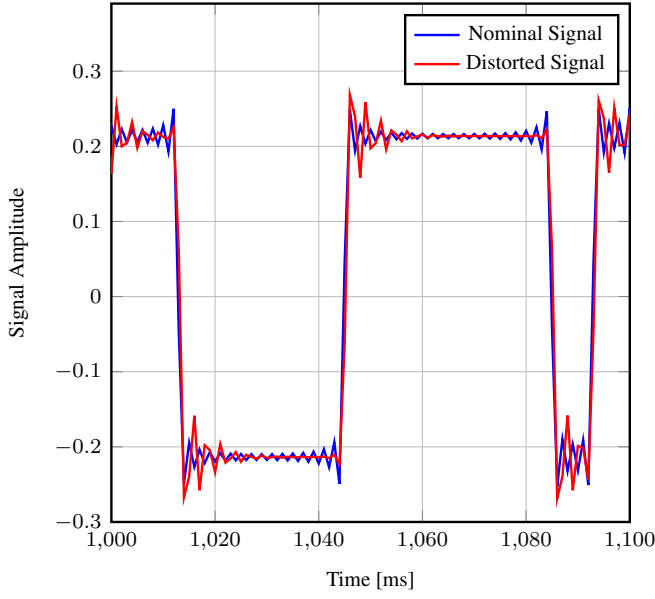


Figure 9. Bandlimited ideal PN signal and analog distorted PN signal.

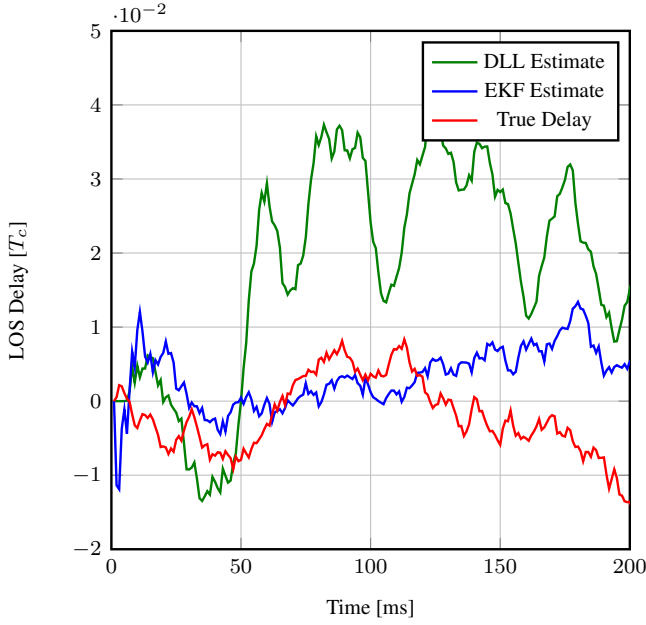


Figure 10. LOS delay estimation comparison between proposed EKF algorithm and conventional DLL for a sudden analog distortion fault.

channel estimation is similar to the one presented in the nominal case simulation. At the moment of the satellite fault however it adapts and the estimated channel coefficients of the proposed algorithm estimate the channel impulse response of the IIR filter with a delay smaller than one T_c .

The performance of that estimation can be seen in Figure 12. In the bar diagram we have plotted the estimated channel coefficients at 100 ms along with the one-sigma standard deviation provided by the EKF versus the true IIR impulse response coefficients with a tab delay smaller than one chip duration T_c . As we can see, the estimation of the coefficients in question matches well their IIR coefficient counterparts.

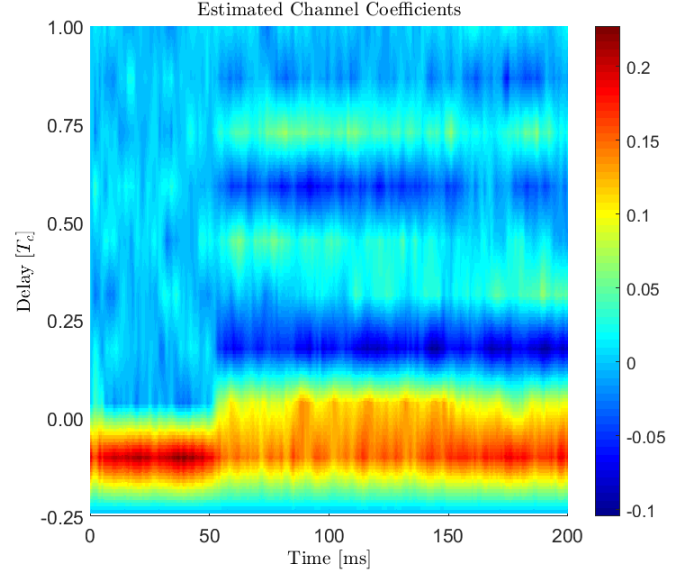


Figure 11. Interpolated channel impulse response for a sudden analog fault.

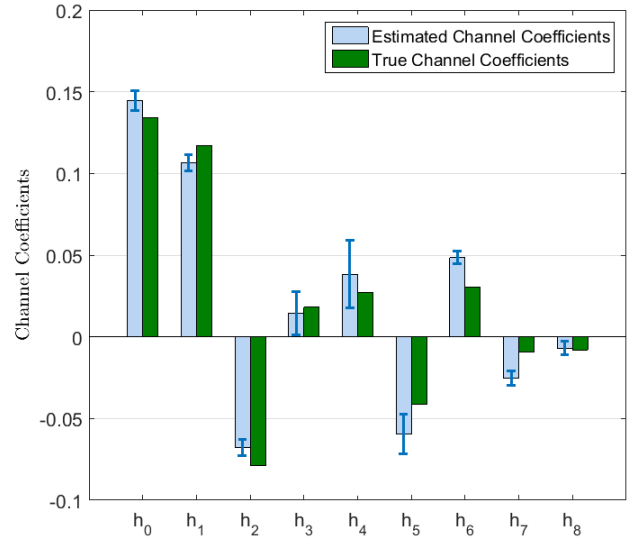


Figure 12. Estimated channel coefficients for the analog distortion fault.

Finally, we compute the signal quality metrics and alarms for the current simulation (Figure 13). During the nominal satellite operation, the signal quality metric lies under the threshold bound. However, as soon as the fault occurs, the magnitude of the metric starts to continuously grow and approximately 7ms after the event, the metric crosses the threshold bound thus triggering a satellite fault alarm.

5. SUMMARY

In this paper, we studied the case of GNSS satellite signal transmission errors. We have analytically derived a model for a GNSS signal channel and analyzed how satellite signal faults can be included in that model. Subsequently, we have

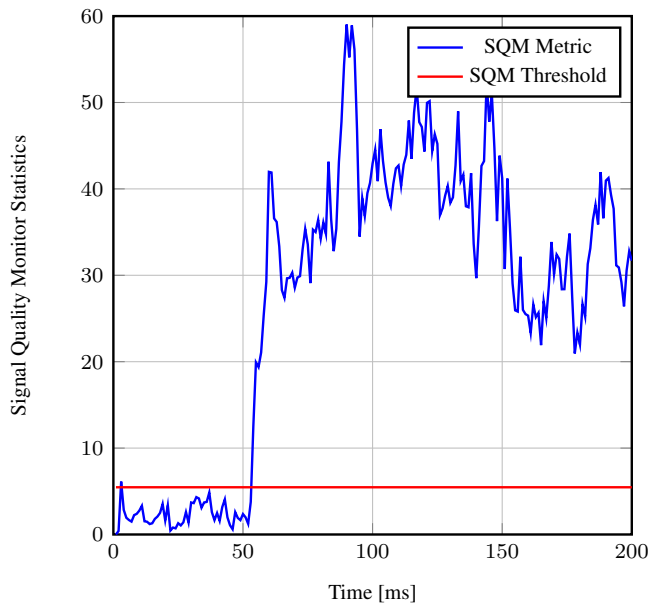


Figure 13. Signal Quality Monitoring (SQM) for sudden analog fault.

developed an Extended Kalman Filter (EKF) algorithm, that is able to estimate the delay of the Line-Of-Signal (LOS) delay but also simultaneously estimate the time-varying channel that the nominal signal experiences. Furthermore, we have compared the performance of the proposed algorithm with a traditional DLL under nominal conditions but also under the influence of a time-varying channel. Finally, in order to test the performance of the proposed algorithm in a GNSS context, where signal transmission distortions are present, we have also modeled the case of sudden analogue distortions on the nominal signal. The proposed algorithm has been subsequently tested under these conditional and its has been shown that it is able to equalize distorted GNSS signals but also provides an alarm metric for the detection of such fault events.

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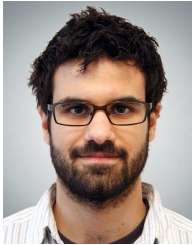
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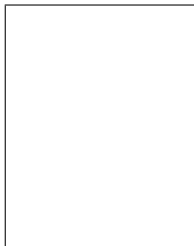
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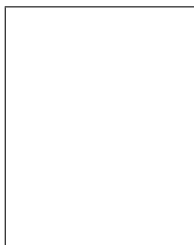
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